

DevOps Series

Ansible Deployment of Jenkins

DevOps, formed by combining the words 'development' and 'operations', is a software development process. Its area of focus is the communication and collaboration between product management, software development and operations professionals. This article is the sixth in the series.

In this article, we will install Jenkins using Ansible and set up a continuous integration (CI) build for a project that uses Git. Jenkins is free and open source automation server software that is used to build, deploy and automate projects. It is written in Java and released under the MIT licence. There are a number of plugins available to integrate Jenkins with other tools such as version control systems, APIs and databases.

Setting it up

A CentOS 6.8 virtual machine (VM) running on KVM will be used for the installation. Internet access should be available from the guest machine. The Ansible version used on the host (Parabola GNU/Linux-libre x86_64) is 2.3.0.0. The *ansible/* folder contains the following files:

```
ansible/inventory/kvm/inventory
ansible/playbooks/configuration/jenkins.yml
ansible/playbooks/admin/uninstall-jenkins.yml
```

The IP address of the guest CentOS 6.8 VM is added to the *inventory* file as shown below:

```
jenkins ansible_host=192.168.122.120 ansible_connection=ssh
ansible_user=root ansible_password=password
```

An entry for the Jenkins host is also added to the */etc/hosts* file as indicated below:

```
192.168.122.120 jenkins
```

Installation

The playbook to install the Jenkins server on the CentOS VM is given below:

```
---
- name: Install Jenkins software
  hosts: jenkins
  gather_facts: true
  become: yes
```



```
become_method: sudo
tags: [jenkins]

tasks:
  - name: Update the software package repository
    yum:
      name: '*'
      update_cache: yes

  - name: Install dependencies
    package:
      name: "{{ item }}"
      state: latest
    with_items:
      - java-1.8.0-openjdk
      - git
      - texlive-latex
      - wget

  - name: Download jenkins repo
    command: wget -O /etc/yum.repos.d/jenkins.repo http://
pkg.jenkins-ci.org/redhat/jenkins.repo

  - name: Import Jenkins CI key
    rpm_key:
      key: http://pkg.jenkins-ci.org/redhat/jenkins-ci.org.
key
      state: present

  - name: Install Jenkins
    package:
```